

Manual

MES Development Suite MLE MDS-MLE 8.1

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1 MLE Customizing - Overview

Fields of Application

The MLE Customizing provides several functions and options to extend existing interface and / or create new interfaces.

Implementation Notes

You use MLE Customizing when:

- You want to change / extend existing interfaces
- Create new interfaces

Integration

The MLE Customizing allows changes and extension in the MLE layer, which is used to provide data for the entire system.

Features

- Several functions and options to extend existing interfaces and / or create new interfaces

2 MES Link Enabling - Overview

2.1 Basics

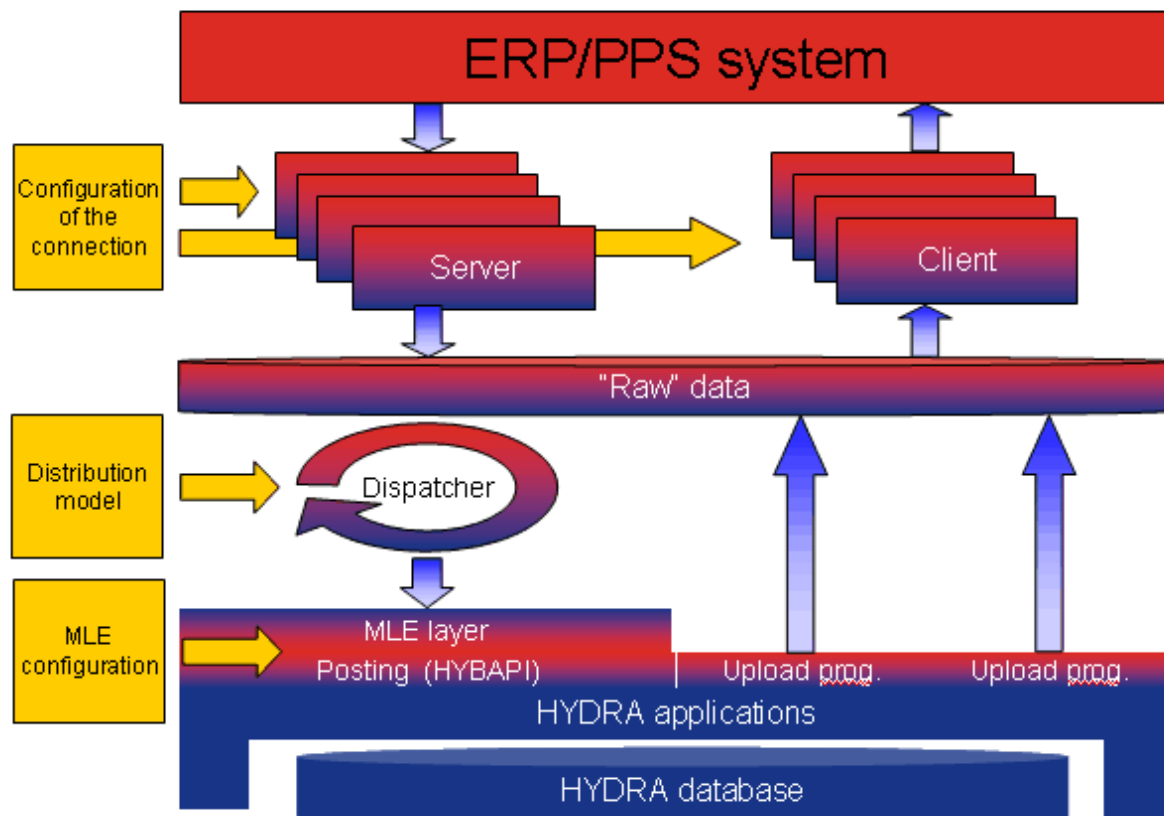
Today modern IT environments rarely consist of so-called “stand-alone” systems. Networking of different systems instead becomes more and more important. A multitude of interfaces, which have to be planned, realized and serviced afterwards, is involved in the networking. At the same time, the requirements with respect to reliability constantly increase while implementation times are reduced.

The HYDRA function MES Link Enabling (MLE) allows for this situation. MLE renders it possible to easily create interfaces from partner systems to transfer data to HYDRA. In this context, MLE works as middleware analyzing incoming data and preparing them in order that they can be transferred to HYDRA via the HYDRA BAPI standard interface and can be posted there. Transferring data via the HYDRA BAPI guarantees that data are processed and posted uniformly in HYDRA, irrespective of the fact whether they have been changed manually in the system or whether they have been transferred by external systems.

This document describes the structure and mode of operation of the MES Link Enabling and shows how and by way of which mechanisms the transferred data can be prepared and formatted.

2.2 System structure

MLE constitutes the middleware between the communication level via RFC/Idoc or file transfer and posting via the HYDRA BAPI. As such, it uses the data provided by the RFC server or file server. It is started from the distribution model of the HYDRA MLE communication. In this context, on the one hand the processing program is started but on the other hand it is also indicated which specific variant is to be used to prepare original data. A variant summarizes the specific rules for transferring data. On the basis of this variant data are prepared into a format that is suitable for HYDRA-BAPI in order to be able to be posted afterwards.



Within the MES Link enabling configuration is based on four levels:

Basic level

The basic level is the highest level of an MLE configuration (of one variant). The definition whether a message type is to be processed transactionally or not is an essential characteristic.

Segment level

One or several BAPIs are assigned to each segment for execution on the segment level. Within the segment level the execution order of the BAPIs can be determined.

Field level

The field level describes which places (from/to) of a data record correspond to which BAPI acronym and thus finally which database field. If the types have to be converted when it comes to the assignment of acronyms they are defined here.

Level of conditions

By way of the condition level fields can be (re)calculated on the basis of original data or their values can be changed depending on certain inspections.

2.3 Integration with other HYDRA modules

Communication with partner systems is configured via the HYDRA MLE communication. For detailed information on the maintenance of these data, please refer to the configuration of MLE communication (hyd-mlek.pdf) document.

The current variants included in the scope of delivery of HYDRA are respectively described in separate documents, for example the documents HKMPP-PDC.pdf as well as HYD-PPS_72.pdf.

When posting data via MLE errors may occur. They can be caused by the posting program, for example, in case the variant indicated within the MLE distribution model is not available. Moreover, errors may appear when posting the data. This can be the case when not all data required for the posting are available or if there are other reasons that are opposed to the data being posted. In these cases, the update program – HYDRA BAPI – displays an error code that defines the posting error. Provided that the escalation management module is licensed, this error code can be forwarded to it and analyzed there.

The formulas used in the level of formulas and conditions are filed in the cross-module formula management.

3 MES Link Enabling - Principles

3.1 Basics of MLE communication

Irrespective of whether it is an SAP Idoc or a file interface, MLE communication is based on a data record consisting of 1000 characters. Depending on the content of the interface in use all 1000 characters may be filled with user data or even less.

Normally, there are no data field labels within these data strings. This means that it is not possible to recognize the type of any place of the data record by merely considering the data record.

The segments or segment names have been designed to differentiate between the different structures within the data string. This name uniquely defines the data string structure. Consequently, statements on the content of a data record can only be made if the segment is known (which segment name).

For this reason, within the scope of MLE segment names control the processing within data packets (IDocs).

3.1.1 BAPI basics

Posting data on the database is based on general guidelines securing consistency and uniformity. Therefore, all writing accesses to the database are carried out by programs providing a unique interface for this purpose.

Consequently, every writing access to the HYDRA database is carried out by programs with a defined interface, irrespective of whether they are started in the HYDRA application or from external applications/systems.

Essentially, this is the HYDRA BAPI. Within the scope of the MES Link Enabling this one is used to post the data transferred to HYDRA from superior systems.

3.1.2 BAPIs and commands

In HYDRA such a BAPI generally exists for each object (that can be maintained via the HYDRA client). Objects in this context may be a (production) order or master data records. Furthermore, there are different methods for each of such objects to access them. In the simplest case it is a method to create (INSERT), change (UPDATE) and delete data records (DELETE).

In more complex cases or if the application requires it further methods are also implemented. This may be modified methods summarizing the insertion and changing or further, application-specific methods.

Such a BAPI is started via the so-called dialog command, which is structured as follows:

<Object>.<Method>

In the following please find an exemplary summary (making no claim on being exhaustive) of available objects and their selected methods.

Object	Methods	Comment
ANR	INSERT UPDATE DELETE MODIFY	The ANR object designates the order.
MNR	INSERT UPDATE DELETE	The MNR object designates machines/workplaces.
FERTVAR	INSERT UPDATE DELETE	The FERTVAR object designates production variants.
RES	INSERT UPDATE DELETE	The RES object designates the resources of the WRM module.

3.1.3 Dialog strings

Having started the BAPI initially via the command user data are transferred in the so-called dialog string or dialog data string. User data are uniquely identified within the dialog string by way of IDs, also known as acronyms.

Such an acronym may represent at least one database field or may have a controlling function in terms of posting purposes. An equals sign "=" and the value transferred for this acronym follows the acronym. The single acronyms and their values are separated by each other as well as from the dialog command by pipes "|".

3.1.4 Customer-specific acronyms/variables

MPDV defines the acronyms for HYDRA standard BAPIs. Moreover, the BAPI processes unknown acronyms without issuing an error message.

In the course of programming an MES Link Enabling variant it could be required to buffer certain values while processing a data record, for example, to be able to recalculate values or to reassign fields subject to conditions.

The MES Link Enabling function does not provide explicit variables for this purpose. However, there is the possibility to benefit from the above-mentioned properties of the HYDRA BAPI. For this purpose, optional acronyms can be used to collect data in a first step. In a second step, they are manipulated and assigned to an acronym via which posting takes place then.

Please note in this context that MPDV cannot guarantee that the acronyms chosen individually will not be interpreted and posted by the HYDRA BAPI at a later point in time. To prevent this it is urgently recommended to use such acronyms only within the customer name space.

The customer name space is identified by the prefix "U:", which means that all acronyms starting with "U:" are protected from being used within and by the HYDRA standard. Moreover, it has to be guaranteed that these acronyms are not utilized within the HYDRA script extensions of the HYDRA standard BAPIs. For this reason, the naming convention

U:MLE_<designation >

should be kept.

In the following please find an example for this procedure:

A single string should be composed of several places of a data record. Only this one should be posted, the individual strings should not be posted.

Acronym	Place from	Place to	Meaning
U:MLE_AKRONYM_1	1	10	Interface fragment
U: MLE_AKRONYM_2	22	23	Interface fragment
U: MLE_AKRONYM_3	34	38	Interface fragment
U: MLE_AKRONYM_4	127	135	Interface fragment
U: MLE_AKRONYM_5	189	195	Interface fragment

Acronym	Place from	Place to	Meaning
ANR.ATKBEZ	0	0	Composite string

The acronyms "U:AKRONYM_1" to "U:AKRONY_5" are created within the customer name space and respectively adopt the interface fragments assigned. The following formula is defined for the "ANR.ATKBEZ" acronym:

U: MLE_AKRONYM_1 + U: MLE_AKRONYM_2 + U: MLE_AKRONYM_3 + U: MLE_AKRONYM_4 + U: MLE_AKRONYM_5

When it comes to posting the composite string of the ANR.ATKBEZ acronym is posted on the database. The acronyms "U: MLE_AKRONYM_1" to "U: MLE_AKRONY_5" are ignored by the BAPI as they are not known to it.

3.2 MLE concepts

The mode of operation of MES Link Enabling provides numerous options to manipulate and edit data transferred from other systems before posting them in HYDRA. A set of basic concepts is used in this context, which are described in the following sections.

3.2.1 Dialog data strings

As already explained, dialog data strings constitute the basis for posting data in HYDRA. MLE provides functions to generate these dialog data strings. A data string consisting of 1000 characters is the basis. The data string can be identified by a segment name.

Within MLE the BAPI used for posting is defined on the basis of the segment name. n-BAPIs can be defined for each segment, whereas the sequence of processing the different BAPIs is defined as well.

Relating to the segment name and the BAPI assigned, it is defined on field level which place of the original data record – identified by "place from" and "place to" – is to be transferred with which acronym to the dialog data string. Prior to the transfer, original data can still be transformed or the transfer might depend on certain conditions. Moreover, it is possible to recalculate values via formulas.

3.2.2 Memory variables

3.2.2.1 General

Customer-specific acronyms render it possible to include data contents in order to use them in formulas, for example. However, this can only be used provided that processing is continued within the same data record. If it is required that the data can still be accessed in the subsequent dialog string, so-called memory variables may be accessed within MES Link Enabling.

These are variables that do not only apply for the processing of a single data record but – based on one data record – are also available for editing other data records.

Among other things, memory variables are always used if, for example, key values of a data record are to be used to identify a subsequent data record in a hierarchical structure. Thus, they have been designed to save data records across all segments. Without them it would not be possible to edit a value of a preceding segment.

Memory variables are configured on field level. It is possible to configure for each field whether it is to be memorized or not (note field). Moreover, it can also be defined whether the value is only to be noted or also to be added to the current dialog string (suppress field).

At run time memory variables are administered in a list that is structured as follows:

Segment name (C30) – Segment name [KEY 1]

Acronym (C40) – Acronym [KEY 2]

Value (C40) – Value

Memory variables may be read in two different ways. On the one hand, there is the direct access when reading a field out (field source = 'M') and on the other hand when using the acronym within a function or condition. When it comes to the processing of formulas and conditions it is always tried at first to fill the variables of the formula/condition from the acronyms of the dialog string. If this fails the list of memory variables is searched. Irrespective of the context in which the memory variables are to be accessed, searching the memory variable list is identical – from back to front.

3.2.2.2 Processing logic

When memory variables are searched for in the list of memory variables it is always searched from back to front, which means, the memory variable list is searched backwards and the first match is processed. Thus, the most current value of an acronym is always in use.

In theory, it might be the case that two acronyms of the same name are saved in the list of memory variables. If this is the case, the acronym inserted at last will be used. If this procedure leads to problems in practice, the indirect way via customer-specific acronyms can be applied. If the value of a customer-specific acronym is then to be assigned to a standard acronym, this can be realized via a respective formula.

Where are the fields of application of memory variables?

As already briefly mentioned, memory variables are used at two spots.

1) Data source:

In general, raw data are cut out of the data segment when editing a field (field configuration: from/to). But there is also the option to configure it in such a way that a memory variable is used instead (field configuration: field source). The acronym that is searched for is also defined within the configuration (field configuration: acronym). If the acronym is found in the list its value is utilized to continue processing.

2) Usage in functions and conditions:

The second option is its use within functions and conditions. In this context, the acronym can be used as if it would be available within the current segment. When it comes later to the search for unknown acronyms, at first the current dialog data string and then the list of memory variables is automatically searched for the acronyms used in the formula.

3.2.3 Formulas/conditions

It is often required to check single field contents with respect to their content or to recalculate them anew or in a different way before they will be booked. In the MES Link Enabling function these requirements are executed on the level of conditions. The formula management of HYDRA MES-Weaver is used as basis in this context.

3.2.3.1 Formulas

On the condition level one or several formulas/conditions can be assigned to a field, which enables individual calculations. The formula itself is defined within the HYDRA MES-Weaver formula management.

The computation of the formula is triggered by transferring the complete formula incl. acronyms, e.g. "ANR.SZY * 1000". The formula is parsed and unknown parameters (acronyms) are determined. Now the triggering routine is provided with them. Having filled the parameters with values the calculation of the formula is restarted and the result is computed.

When the values for the parsed acronyms are determined it is proceeded as follows:

At first, it is checked whether it is the current acronym of the field which is assigned to the formula. In this case, the current value would be entered.

If it is not the current acronym the DLG string generated so far will be searched for it.

If this fails as well the memory variables will be searched.

If this search is not successful either an error will be forwarded to ESK and the values will not be taken over. (Processing of this field will be cancelled.)

If the acronym can be found as described above it will be used as new input value of the calculation routine and the next unknown acronym will be searched.

Provided that another step is defined within the configuration of formulas and conditions this one will now be carried out. But now the newly computed result is to be entered. If no further step follows the result is assigned to the acronym.

Please note: A formula may only contain acronyms, which have already been entered in this BAPI dialog, are currently being edited or are available within the memory variable list. If this is not the case, the order of filling the DLG string has possibly to be changed.

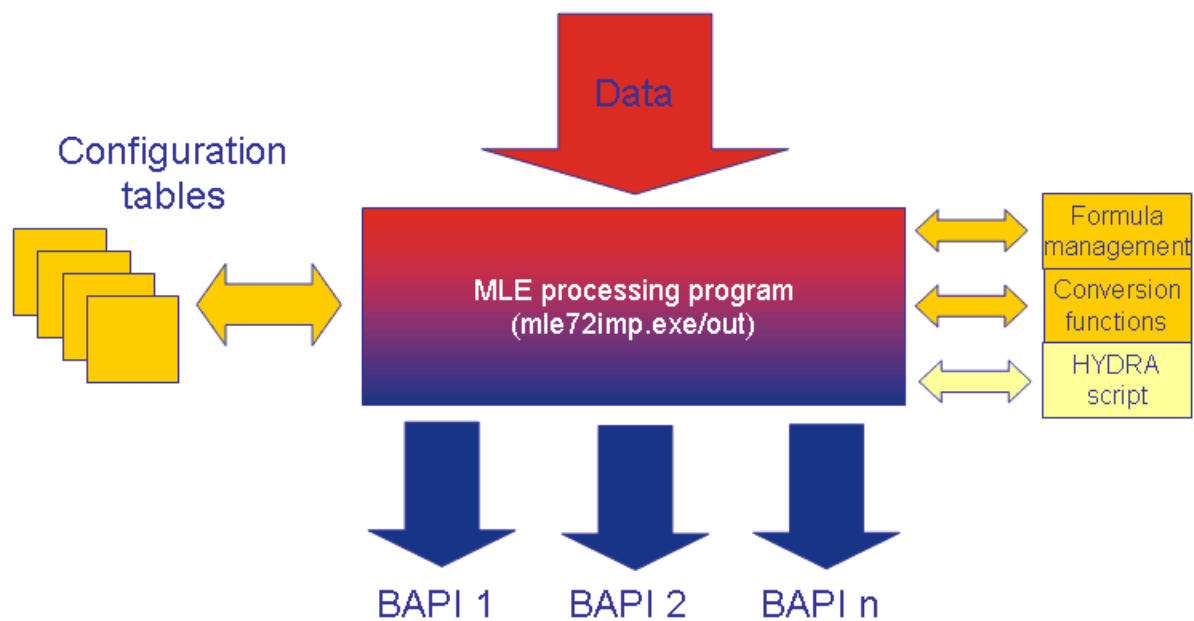
3.2.3.2 Conditions

Conditions are checked along the lines of calculation functions. Unknown acronyms are found and are made available for the triggering routine to enter real values. A condition is identified by the "B" ID within the configuration.

In contrast to formulas, conditions, however, do not return a computed result but the values "TRUE" (the condition is true) or "FALSE" / "0" (the condition is false). Depending on the inspection result the further processing may be defined. In this context, it is differentiated whether the result of the condition turned out positive or negative.

3.3 System structure

MLE ranks between the communication level (below) and the posting level (above). Basically, MLE consists of a processing program. This program analyzes the configuration defined in the database. In addition to this, it accesses other components of the HYDRA MES-Weaver.



3.4 Namespaces

To protect the customer application from being overwritten unintentionally by MPDV, MLE differentiates between MPDV namespace and customer name space. The namespaces are distinguished by different initial letters of the MLE variant.

Customer name space

Variants of the customer name space always start with "U:", which is guaranteed by the system.

MPDV name space

MPDV can use all other letters and numbers to designate variants. MPDV does never deliver a standard variant to the customer name space.

4 MLE Basic Configuration

Overview

Menu	System administration → MES Link Enabling (MLE) → MLE basic configuration
Transaction code	mlebcfg
Function authorization	mlebcfg

Purpose

You use the application to create or change MLE variants in the system.

Integration

MLE variants are combinations of configurations used to process data in HYDRA inbound processing.

Requirements



Some of the functions require the development license MDS-MLE to be fully available. Without development license, you can in general only display the data, but not change the data. The functions, which are only available with development license, are identified via (*).

Field descriptions

Message type

The message type is the central key in MLE inbound processing. With SAP R/3, the type is specified by SAP. If other PPS systems are used, it is defined by HYDRA.

In the inbound processing, the MLE dispatcher uses the message type of the distribution model to identify the current variant for this message type. The MLE Dispatcher then calls the processing program with the message type and the variant.

Variant

The variant defines a specific configuration for a message type. A variant includes all relevant settings for a specific message type. To activate a variant, the variant is stored in the MLE distribution model. For each message type, different variants may be available.

By default, HYDRA includes several variants for specific interfaces. These variants cannot be changed in the MLE configuration. If you want to change a variant according to your requirements, this variant must be copied to the customer namespace. This namespace starts with "U:". The system rejects changes in the MPDV namespace.

Message function

You use the message function to control the processing in detail. Using the message function, you can externally identify the processing of data records without having to define different segments.

The message function is transferred to HYDRA as follows:

If the partner system is an SAP system, the message function is transferred with the IDoc from SAP.

If the partner system is a PPS system, the extension of the file provided by the PPS system identifies the message function. The documentation of the MLE Communication (HYD-MLEK.pdf) lists the message functions that are supported by MLE.

Processing

The processing defines whether the program formats data using the configuration and transfers it to HYDRA or whether the complete processing is exported to a user exit.

Edit configuration:

The processing program takes all the data required for the transfer from the MLE configuration.

User exit

If the *User exit* (*mle_verarbseg_in.hsc*) is used on the basic level, any MLE configuration is ignored and the user exit exclusively controls the data transfer.

The data records are transferred row by row from the inbound table (*hysap_inbound_data*) to the user exit. Also the parameters message type (MESTYP), message function (MESFCT) and variant are transferred. The user exit processes the rows using the segment name (SEGNAM) and writes the result in up to three return strings. The number of possible BAPI strings is therefore limited to 3 strings that are transferred to the server one by one.

NOTE:

It is not enough to set the option to *User exit* to implement the user exit. Further steps are required.

Comment

You can use the comment to enter notes as free text.

Transaction

You use the transaction to control the transfer logic. This means that you can configure if an IDoc (a transaction) is only transferred as a whole or if also parts can be transferred.

If an IDoc with transaction = "J" is transferred, the data records are only actually posted, if ALL data records of the IDoc have been posted successfully. If this fails for only one data record, NO data record of the IDoc is transferred.

For this reason it is recommended to do without transactional brackets with interfaces based on operations (HY72PPS / PP-PDC). With the transfer of master data, it can be useful to transfer all data records or none.

Toolbar



MLE Segment configuration

You can show the [Segment configuration](#) for a selected data record.

Insert, Edit, Delete (*)

Use these functions to edit data records.



Copy (*)

Use this function to copy an existing data record and create a new MLE variant in the system using this copy.

You can only copy the header of the selected basic configuration or the complete configuration including its structure.

5 MLE Segment Configuration

Overview

Menu	System administration → MES Link Enabling → MLE segment configuration
Transaction code	mlescfcg
Function authorization	mlescfcg

Purpose

The segment configuration is the configuration level used to store the BAPIs (e.g. ANR.MODIFY) that are executed for the specified segment.

Integration

MLE variants are combinations of configurations used to process data in HYDRA inbound processing.

Requirements



Some of the functions require the development license MDS-MLE to be fully available. Without development license, you can in general only display the data, but not change the data. The functions, which are only available with development license, are identified via (*).

Field descriptions

Message type

The message type refers to a message type created in the basic configuration.

Variant

The variant refers to a variant created in the basic configuration.

Message function

The message function refers to a message function created in the basic configuration.

Segment

The segment is the key on segment level. For each segment of an IDoc, you specify the BAPIs that are called for this segment. The segment name must be identical to the segment name of the IDoc.

Hierarchy

A hierarchy level is assigned to each segment. The value "1" is automatically preassigned. This is the highest hierarchy level of an IDoc. You can assign more than one segment to each hierarchy level.

You use the hierarchy level to specify the structures in an IDoc. This controls the processing. It is therefore possible that specific segments are not transferred if the transfer in a higher hierarchy level was not successful. Example:

If the system identifies for an operation that data of this operation cannot be transferred because the operation is running, then the transfer of segments that belong to this operation (e.g. components) can be skipped.

To this end, you must combine the assignment of hierarchy levels and the definition of the segment type "Special handling of OP".

Processing

The processing defines whether the program formats data using the configuration and transfers it to HYDRA or whether the complete processing is exported to a user exit.

Edit configuration:

The processing program gets all data required for the transfer from the MLE configuration.

User exit

On segment level, three user exits can be used. Each user exit intervenes at a specified time in the transfer process:

User exit before MLE processing (UB):

On segment level, the user exit (*mle_modifysapdata_in.hsc*) provides a complete SDATA string. Further parameters that are transferred are message type (MESTYP), message function (MESFCT) and variant. The user exit can then prepare the string. Afterwards the string runs through the configured MLE process.

User exit instead of MLE processing (U):

If the user exit (*mle_convsapdata_in.hsc*) is used, the data records are transferred row by row to the user exit. In return, BAPI strings are expected that are directly posted. Processing via MLE configuration does not take place.

User exit after MLE processing (UA):

If the user exit (*mle_modifydlgstr_in.hsc*) is used, a BAPI dialog string created by the MLE configuration is transferred to the user exit. This dialog string can be modified in the user exit. The dialog string is then transferred to the BAPI for posting. MLE does not perform a further check and/or processing.

NOTE:

It is not enough to set the option to *User exit* to implement the user exit. Further steps are required.

Call BAPI directly (B):

You can also transfer data via MLE interface if the data is directly available in HYDRA dialog data format. An interpretation in the MLE layer is not required. In this case, you can set the flag *Call BAPI directly* on segment level. The segment definition requires the following structure:

Field name	Type		Description	Example
Transaction	CHAR	20	Transaction ID (dialog ID in HYDRA)	PNR.MODIFY
Description	CHAR	40	Plain text designation as comment	Download HR master
Data	CHAR	940	Dialog data string for HYDRA	DLG=PNR.MODIFY PNR.PNR=12345678 PNR.KNR=0000001 Details see section 2.3.1

If the flag is set on segment level, no further evaluation / processing of existing configurations is performed on the field or condition level.

BAPI dialog

The BAPI dialog describes the object (order, component,...) and the kind of posting made for the object.

To store the BAPI, the following pattern is used: <object>.<method>. Example: "ANR.MODIFY" to modify order data.

Segment sorting

You can use the segment sorting to define a specific sequence of segments. The sorting is not fixed. You can change the sorting at a later time.

You can manually define the segment sorting. Just enter the respective number. If you do not change the default value ("0") in the field, the system automatically identifies the next consecutive number. To identify the next consecutive number, the system uses steps of 10.

Currently, the segment sorting does not include a function.

BAPI sorting

You can use the BAPI sorting to define a sequence for the different BAPI calls that are filled with the data of a segment. This is required if you want to transfer the order header and the operations from a segment, for example. You can change the defined sorting at any time.

You can manually specify the BAPI sorting. Just enter the respective number. If you do not change the default value ("0") in the field, the system automatically identifies the next consecutive number. To identify the next consecutive number, the system uses steps of 10.

The MLE processing program prepares the dialog strings for the BAPIs using the defined BAPI sorting and posts the strings in the defined order.

Consec. BAPI no.

The consecutive BAPI number can be assigned manually or automatically by the system. This number internally identifies a data record.

It is recommended to assign the number automatically. In this case, you must not change the default value ("0").

Segment type

The segment type is used to control the transfer.

Segment type " " (default):

The segment is processed according to the MLE configuration.

Segment type "Special handling of OP" (AS):

If you use the segment type *Special handling of OP*, the processing of the subsequent hierarchy levels depends on the processing result of the currently configured segment.

As a result, the segments of a lower hierarchy level are not executed if the segment of a higher hierarchy level could be posted. Example:

If the system finds out that you cannot make postings for an operation (hierarchy level 2) because it is running, then also for the components (hierarchy level 3) you cannot make postings. The system cancels the processing of these segments. The next data record that is executed, is a segment of the same level or of a higher (e.g. "1") hierarchy level than the operation.

BAPI call

The *BAPI call* defines how often a BAPI of a segment can be called.

Always run BAPI (A):

The BAPI is executed with every segment processing. How often a BAPI is executed is therefore specified by the number of segments of this segment type per IDoc.

Run BAPI only once (O):

The BAPI is executed once per segment AND per IDoc. The number of BAPI runs per IDoc does therefore not depend on the number of segments.

Delete flag

The memory variables are completely deleted if a segment of the same name is available a second time.

☐ Here, the memory variables of the segment are not deleted, but overwritten by new values.

Dialog data

In the dialog data, you can store fixed acronyms that are transferred with each BAPI call. Close each acronym using "|". To store fixed dialog data, 100 characters are available.

If this is not enough, you can store further acronyms on field level.

Toolbar



MLE field configuration

You can show the [Field configuration](#) for a selected data record.

Insert, Copy, Edit, Delete (*)

Use these functions to edit data records.

6 MLE Field Configuration

Overview

Menu	System administration → MES Link Enabling → MLE field configuration
Transaction code	mlefcfg
Function authorization	mlefcfg

Purpose

On field level, you specify the section of the data record that is assigned to a specific BAPI acronym. This way, also the database field is specified where the data string section is stored.

Before the assignment to a BAPI acronym, you can convert the data to the HYDRA data format. The system provides a series of conversion routines to this end.

Integration

MLE variants are combinations of configurations used to process data in HYDRA inbound processing.

Requirements



Some of the functions require the development license MDS-MLE to be fully available. Without development license, you can in general only display the data, but not change the data. The functions, which are only available with development license, are identified via (*).

Field descriptions

Message type

The message type refers to a message type created in the segment configuration.

Variant

The variant refers to a variant created in the segment configuration.

Message function

The message function refers to a message function created in the segment configuration.

Segment

The segment refers to a segment created in the segment configuration.

Consec. BAPI no.

The consecutive BAPI number links the field configuration to the segment configuration. Because it is possible to call several BAPI calls per segment, the number is used to link to exactly one BAPI call.

The consecutive BAPI number explicitly refers to the static, consecutive BAPI number and NOT to BAPI sorting.

You must assign this value with the first field configuration of a segment. With all further field configurations, this number is used as default value if an existing field configuration is selected.

Consec. field no.

The consecutive field number can be assigned manually or automatically by the system. This number internally identifies a data record.

It is recommended to assign the number automatically. In this case, you must not change the default value ("0").

Sort sequence

You use the sort sequence to build a BAPI dialog string in a specific sequence. The sequence specifies the sections of the data string that are first cut and inserted in the dialog string. The sections are chronologically inserted in the specified sequence from left to right.

The sequence used to insert data into the dialog string is important if the values of a dialog string are offset or compared using formulas. When the formulas are processed (level: condition configuration), the variables used in the formula are searched in the dialog string from right to left (i.e. the reverse chronological order of generation). For this reason, it is useful to cut sections that are relevant for the formulas and to include them in the dialog string.

You can manually specify the sort sequence. Enter the respective number. If you do not change the default value ("0") in the field, the system automatically identifies the next consecutive number. When the next consecutive number is identified, the system uses steps of 10. You can change the sort order at any time.

Field designation

The field name refers to a section of the data string. The name stored is displayed as field name in the MLE communication (menu: *File --> System information --> MLE communication --> Inbound/outbound transactions*).

Acronym

The acronym uniquely defines a section of a data record and therefore defines the database field where the data is stored. The acronym is stored without "=".

From/to

The fields *From* / *To* specify the position in the data string where the required data is placed. The right and the left limits are specified. Example:

From: 4

To: 7

Data record: ABCDEFGHIJKLMNO

The data positions 4, 5, 6 and 7 are used. Result: "DEFG" are inserted in the dialog data string.

If you want to insert acronyms in the dialog string that cannot directly be cut in a data string or that are calculated using a formula, then you must enter "0" in both fields.

Function

You use the function to convert data from the data string before the data is inserted in the dialog string. This is required, e.g. if time or date formats do not match the HYDRA internal format. MLE provides the required routines:

SAP QUAN --> HYDRA DEC (FKT02):

The function converts data from the SAP QUAN format ("nnnnnnnnnn.mmm+") to the HYDRA internal DEC format ("+nnnnnnnnnn.mmm").

The function also converts data from the exponential format of the PP-PI-PCS interface (e.g. 2.3650000000000002E+00) to the HYDRA decimal format.

SAP DATE --> HYDRA DATE (FKT03):

The function converts data from the SAP date format "YYYYMMDD" to the HYDRA date format "MM/DD/YYYY".

SAP TIME --> HYDRA TIME (FKT04):

The function converts data from the SAP time format "HHMMSS" to the HYDRA internal time format seconds since midnight.

HY71PPS date → HYDRA date (FKT05)

The function converts data from the 8 digit date format of the HY71PPS interface to the HYDRA date format.

HY71PPS (DAT6) Date → HYDRA Date (FKT06)

The function converts data from the 6 digit date format of the HY71PPS interface to the HYDRA date format.

To Upper --> Upper case letters (FKT07)

The function converts all characters of a string to UPPER CASE LETTERS.

To Lower --> Lower case letters (FKT08)

The function converts all characters of a string to lower case letters.

Sprintf() Funktion --> Conversion of input values (FKT09)

You can use this function to convert types.

SAP QUAN --> HYDRA DEC round (FKT10)

Similar to the function FKT02, this function converts values. But this function also rounds to integer values.

Parameter

Parameters are filled automatically if one of the above-mentioned conversion functions is used. It is not recommended to make manual entries.

Processing

The processing defines whether the program formats data using the configuration and then transfers the data to HYDRA or whether the processing for the relevant field is exported to a user exit.

Edit configuration:

The processing program takes all the data required for the transfer from the MLE configuration.

User exit

If the user exit (*mle_convfield_in.hsc*) is used, the user exit is provided with the original field content of the data string. The user exit then provides the prepared value that can directly be transferred into the dialog string by MLE without further processing.

Comment

You can store comments in this field.

Field type

This is the field type specified for the MLE processing. By default, all fields are text fields. Here, the conversion routines stored with *Function* can also be used.

The field type is important if the contents of several acronyms are combined in string operations (via the condition level). By default, the function stored identifies independently if it is a string or a figure. With data fields that are strings and include numbers, this leads to errors.

If the field type is specified, you can control this. You can then also combine figures in strings.

Field source

The field source specifies the data basis used to insert data in the dialog string. Several options are available:

Data string (default):

In this case, data is taken from the current, original data string. This is the "normal case".

Constant value:

If a constant value is used, this value is integrated in the dialog string. The value itself is defined in the *Field data* field.

Difference segment configuration --> dialog data

Acronyms and their values stored in the dialog data are in any case integrated in the dialog string. To integrate the constant stored in the dialog string, you can additionally define a condition.

Memory variable:

Memory variables are used to receive specific field contents of a data record and to make them available to other data records.

If you specify a field as memory variable, the system tries to access the acronym of the memory variable stored under *Field data*. If the memory variable is found, the value of the memory variable is included in the dialog string.

Field data

Field data is used in two different cases – if *constant value* or *memory variable* is defined as the *Field source*.

Constant value:

The field includes the value of the constant. The acronym has already been defined within the *Acronym* field in the *General* tab.

Memory variable:

The field contains the acronym of the memory variable. The system searches the memory variable in the existing pool of memory variables. The value of the memory variable and the value defined in the *Acronym* field (*General* tab) are integrated in the current dialog string.

Suppress acronym

If this option is enabled, the acronym and the value from the current data string are included in the list of memory variables and are then available in the subsequent segments. The value is not integrated in the current dialog string. This option is only useful in combination with the option *Memorize acronym*.

If this option is not active, the acronym and the value from the current data string are included in the list of memory variables and are then available in the subsequent segments. The acronym and the value of the current data string are additionally integrated in the current dialog string.

Memorize acronym

If this option is enabled, the acronym and the value from the current data string are included in the list of memory variables and are then available in the subsequent segments. The option *Suppress acronym* specifies if the acronym is additionally integrated in the current dialog string.

If the option is not active, the acronym and the value from the current data string are not included in the list of memory variables and are then not available in the subsequent segments. The acronym and the value of the current data string are only integrated in the current dialog string.

Display in MLE communication

If the option is enabled, the field is displayed in the MLE communication with the field designation stored.

If the option is not enabled, the field is not displayed in the MLE communication with the field designation stored.

Toolbar



MLE formula/conditions

You can show the [Formula/Condition configuration](#) for a selected data record.



Field up

Moves a field in the field sorting one level up.



Field down

Moves a field in the field sorting one level down.

Insert, Copy, Edit, Delete (*)

Use these functions to edit data records.

7 MLE Formula/Conditions

Overview

Menu	Systemadministration → MES Link Enabling (MLE) → MLE formula/conditions
Transaction code	mleecfg
Function authorization	mleecfg

Purpose

The configuration of conditions provides the following possibilities:

- You can check conditions before you transfer data to HYDRA and if required you can then change the data.
- You can influence the further processing.
- You can use formulas to recalculate the data transferred.
- You can use formulas to calculate new data using the data transferred.

Integration

MLE variants are combinations of configurations used to process data in HYDRA inbound processing.

Requirements



Some of the functions require the development license MDS-MLE to be fully available. Without development license, you can in general only display the data, but not change the data. The functions, which are only available with development license, are identified via (*).

Field descriptions

Message type

The message type refers to a message type created in the segment configuration.

Variant

The variant refers to a variant created in the segment configuration.

Message function

The message function refers to a message function created in the segment configuration.

Segment

The segment refers to a segment created in the segment configuration.

Consec. BAPI no.

The consecutive BAPI number references a specific BAPI of a segment.

You must assign this value with the first configuration of a formula / condition for a field. With all further configurations, this number is used as default value if an existing configuration is selected.

Consec. field no.

The consecutive field number references a specific field. The formula or condition is assigned to this field.

The consecutive field number explicitly refers to the static, consecutive field number and NOT to the field sorting.

You must assign this value with the first configuration of a formula / condition for a field. With all further field configurations, this number is used as default value if an existing field configuration is selected.

Consec. formula no.

The consecutive formula number can be assigned manually or automatically by the system. This number internally identifies a data record.

It is recommended to assign the number automatically. In this case, you must not change the default value ("0").

Sort sequence

You use the sort sequence to calculate or check the formulas/conditions in a specified sequence. Using the sort sequence, the conditions are checked one after the other. The processing of the sort sequence can directly be controlled via the fields *With neg. result* / *With pos. result*.

You can manually specify the sort sequence. Directly enter the respective number. If you do not change the default value ("0") in the field, the system automatically identifies the next consecutive number. When the next consecutive number is identified, the system uses steps of 10. You can change the sort sequence at any time.

Type

The type specifies if it is a formula or a condition.

Formula:

Formulas return a calculated value as result.

Condition:

Conditions return the values *True* or *False* as result. The relevant value can then trigger the processing of further conditions or formulas.

Formula

Here, you can access the global [Formula management](#). If the required formula does not exist, you can create the formula.

To classify the formulas stored in the system, you use the formula type. We recommend to use the formula type *50 MLE conditions (MES Link Enabling)* for formulas used here.

You can use a placeholder when you define formulas. The placeholder is replaced with the current acronym before the formula/condition is checked. This way, you can use the same formula/condition for different fields.

Use %AKR% as placeholder in the formula.

Result with pos. condition

The result with positive condition is relevant for formulas and conditions.

Formula:

If formulas are used, the calculated value can be integrated using "%ERG%" The string "%ERG%" is then replaced with the calculated result of the formula.

NOTE:

In this row, only the data itself may be entered. The acronym is already specified via the field referenced.

Condition:

In case of a condition, three options are possible:

"%ERG%":

Here, the result of the condition is used, i.e. "0" (false) or "1" (true).

"%AKR%"

Here, the original value of the acronym is used.

Value

Instead of the integrated or calculated values, you can also store a new value.

Result with neg. condition

The result with negative condition is normally only relevant with conditions.

Three options are generally possible:

"%ERG%":

Here, the result of the condition is used, i.e. "0" (false) or "1" (true).

"%AKR%"

Here, the original value of the acronym is used.

Value

Instead of the integrated or calculated values, you can also store a new value.

With neg. condition / With pos. condition

Using these settings, you can combine several formulas or conditions. This way, you can check a complete cascade of conditions. Important: Independent of the number of values checked, only one value/result can be integrated in the dialog string.

Depending on the result, the following processing options are available:

"W" – Continue:

If the option "W" is selected, the system continues to process the formulas/conditions stored next. The sort sequence specifies the formula/condition that is checked next.

"S" - Stop:

If the option "S" is selected, the processing of formulas/conditions is stopped here. The system uses the value stored in the relevant result row (%ERG%, %AKR%, value) as value in the acronym and dialog string.

"ES" – Exit Segment:

If the option "ES" is selected, the generation of the dialog data string is canceled. This means that all existing data of the current dialog string is deleted and the next segment is processed. The hierarchy level of the subsequent segment is not important in this context.

"EH" – Exit hierarchy:

If the option "EH" is selected, the generation of the dialog string is canceled. This means that all existing data of the current dialog string is deleted. The processing is continued with the next segment of a higher hierarchy level. All segments are skipped, which are of the same or of a lower hierarchy level between the canceled segment and the next segment of a higher hierarchy level

"EM" – Exit Minor:

If the option "EM" is selected, the generation of the dialog string is canceled. This means that all existing data of the current dialog string is deleted. The processing is continued with the next segment of a same or a higher hierarchy level. All segments are skipped, which are of a lower hierarchy level between the canceled segment and the next segment of a same or a higher hierarchy level

"EB" – Exit Bapi:

If the option "EB" is selected, the generation of the current dialog string is canceled. This means that all existing data of the current dialog string is deleted and the dialog string is not executed. The processing is continued with the next BAPI of this segment or the next segment. The system does not set a failure status or similar. The current BAPI dialog is only skipped. A relevant logging in the inbound transactions is made.

Comment

You can use this field to store a comment.

Toolbar**Insert, Copy, Edit, Delete (*)**

Use these functions to edit data records.

8 SAP Order Sequencing

Overview

HYDRA menu	System administration → MES Link Enabling → SAP order sequencing
FEDRA menu	System administration → MES Link Enabling → SAP order sequencing
Transaction code	mleoss
Function authorization	mleoss.*

Purpose

Use the *SAP order sequencing* to control how the workplaces specified by SAP are transferred to the system. You can choose from the following options:

- The workplace transferred from SAP is interpreted as HYDRA group and the operation is planned in the pool of groups (backlog for machine group).
- The HYDRA group is selected for the workplace transferred from SAP and the operation is planned in the pool of groups (backlog for machine group).
- The HYDRA group is selected for the workplace transferred from SAP and the operation is directly planned for the machine.

This decision either affects the workplace or the entire system.

Integration

Diverse interfaces use these configurations to transfer orders from SAP.

Requirements

You have created workplaces and groups in the system.

Field descriptions

Key

Use this field to specify whether the entry applies to a specific order type or a machine.

In general, the configuration refers to a workplace.

Value

If you selected *order type* as the key, use this field to enter the order type the configuration applies for.

If you selected *machine* as the key:

- Enter a separate workplace in the *value* field if the configuration should apply for a specific workplace.
- Enter the value SYSTEM if you want the configuration to apply for the entire system.



Configurations referring to a specific machine take priority over the SYSTEM setting.

You can make a system entry for the majority of machines/workplaces/work centers and exceptions may be configured specifically.

Detailed planning

- **G**

"Transfer the SAP workplace as HYDRA group, plan operation in the pool of groups (backlog for machine group)."

- **M**

"Transfer the SAP workplace as HYDRA workplace, identify the HYDRA group, plan operation in the pool of groups (backlog for machine group)."

- **N**

"Transfer the SAP workplace as HYDRA workplace, identify the HYDRA group, and plan the operation for the workplace."

- **Rule 1 - Rule 9**

Use the rules 1 to 9 for customer-specific transfer logics, which will be implemented as part of the project.

9 SAP Upload

Overview

HYDRA menu	System administration → MES Link Enabling (MLE) → SAP upload
FEDRA menu	System administration → MES Link Enabling (MLE) → SAP upload
Transaction code	mlecos
Function authorization	mlecos.*

Purpose

You use this function to configure the transfer from the resource performance accounts to the SAP activity types. You can make this setting for a single workplace or globally for the entire system.

Integration

This setting is used when the SAP time tickets are uploaded to SAP PP via the PP-PDC interface.

Requirements

You have created workplaces and groups in the system.

Field descriptions

Workplace

If a machine is specified, the configuration is applied for this machine (system workplace). If you enter "SYSTEM", this is a global setting and is used for all machines, except for machine with an own, specific entry.

Activity 1 – Activity 6

These fields specify which combination of time accounts kept in the system is uploaded. Possible values:

- RPA1 – RPA12
Resource performance accounts of machine from U/E record
- PRPA1 – PRPA12
Resource performance accounts of persons from U/E record (the upload of times from the resource performance accounts of persons [must be activated separately](#).)
- P_DAUER
Labor utilization

To post the total of several time accounts to an activity field, the accounts are listed separated by the "+" sign.

Example: RPA1+RPA2+RPA3+RPA4+RPA5+RPA6+RPA7+RPA8+RPA9+RPA10+RPA11

You can also upload quantity accounts, not only time accounts. Possible values are:

GUT (yield),

AUS (scrap).

To post the total of several quantity accounts to an activity field, the accounts are listed separated by the "+" sign.

Example: GUT+AUS

Unit 1 – Unit 6

Enter the relevant upload unit in these fields. Valid values:

Hours	H, HUR, HR, STD
Minutes	MIN
Seconds	SEC
One tenth of an hour (6 min)	ZE

Target activity 1 – Target activity 6

If you check this option, you activate the calculation of the activities using the number of pieces defined in SAP for the activity type. If the option is enabled, the activities are no longer transferred to SAP. If you disable the option, the transfer is enabled again. Also this setting can be made globally for the entire system or for a specific machine only. If the fields are checked, the fields TARGET_ACTI1 – TARGET_ACTI6 in the upload structure "TIMETICKET" are equally assigned an "X".

10 Activity Types SAP

Overview

HYDRA menu	System administration → MES Link Enabling (MLE) → Activity types SAP
FEDRA menu	System administration → MES Link Enabling (MLE) → Activity types SAP
Transaktionscode	mlecas
Function authorization	mlecas.*

Purpose

You use this setting if you want to link the activity types required for uploads to SAP PM with the workplaces created in the system.

You edit the assignment in relation to a workplace known in the system and the target module in SAP.

Integration

The setting is used for uploads to SAP PM.

Requirements

You have created workplaces and groups in the system.

Field descriptions

Activity type

Activity type to stipulate how to upload to SAP.

Cost center

Not relevant

SAP_Modul

Enter "PM" permanently for uploads to SAP PM.

Name

Free text

Year

No functional use: Date of validity

Plant

No functional use: Plant where the workplace is located

Workplace

Workplace to which the setting applies

From PPS

Don't set the identification.